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Industrial IoT and Industry 4.0

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Installation of Node-Red in Windows

❏ Node.js Installation

Download the Node.js Installer

- Go to the official Node.js website: <https://nodejs.org>
- You will see two versions:
 - **LTS (Recommended for Most Users):** This is the Long Term Support version. It is the most stable and reliable. **You should download this one.**
 - **Current:** This is the latest version with the newest features, but it may be less stable. It's for users who need the latest updates.
- Click on the **LTS** button to download the Windows Installer (.msi file). The download will start automatically.

Download Node.js®

Get Node.js®

v22.19.0 (LTS) ▾

for

🇺🇸 Windows ▾

using

🐳 Docker ▾

with

📦 npm ▾

Info Want new features sooner? Get the [latest Node.js version](#) instead and try the latest improvements!

```
1 # Docker has specific installation instructions for each operating system.
2 # Please refer to the official documentation at https://docker.com/get-started/
3
4 # Pull the Node.js Docker image:
5 docker pull node:22-alpine
6
7 # Create a Node.js container and start a Shell session:
8 docker run -it --rm --entrypoint sh node:22-alpine
9
10 # Verify the Node.js version:
11 node -v # Should print "v22.19.0".
12
13 # Verify npm version:
14 npm -v # Should print "10.9.3".
```

PowerShell

 Copy to clipboardDocker is a containerization platform. If you encounter any issues please visit [Docker's website](#) ↗

Or get a prebuilt Node.js® for

🇺🇸 Windows ▾

running a

x64 ▾

architecture.



Windows Installer (.msi)



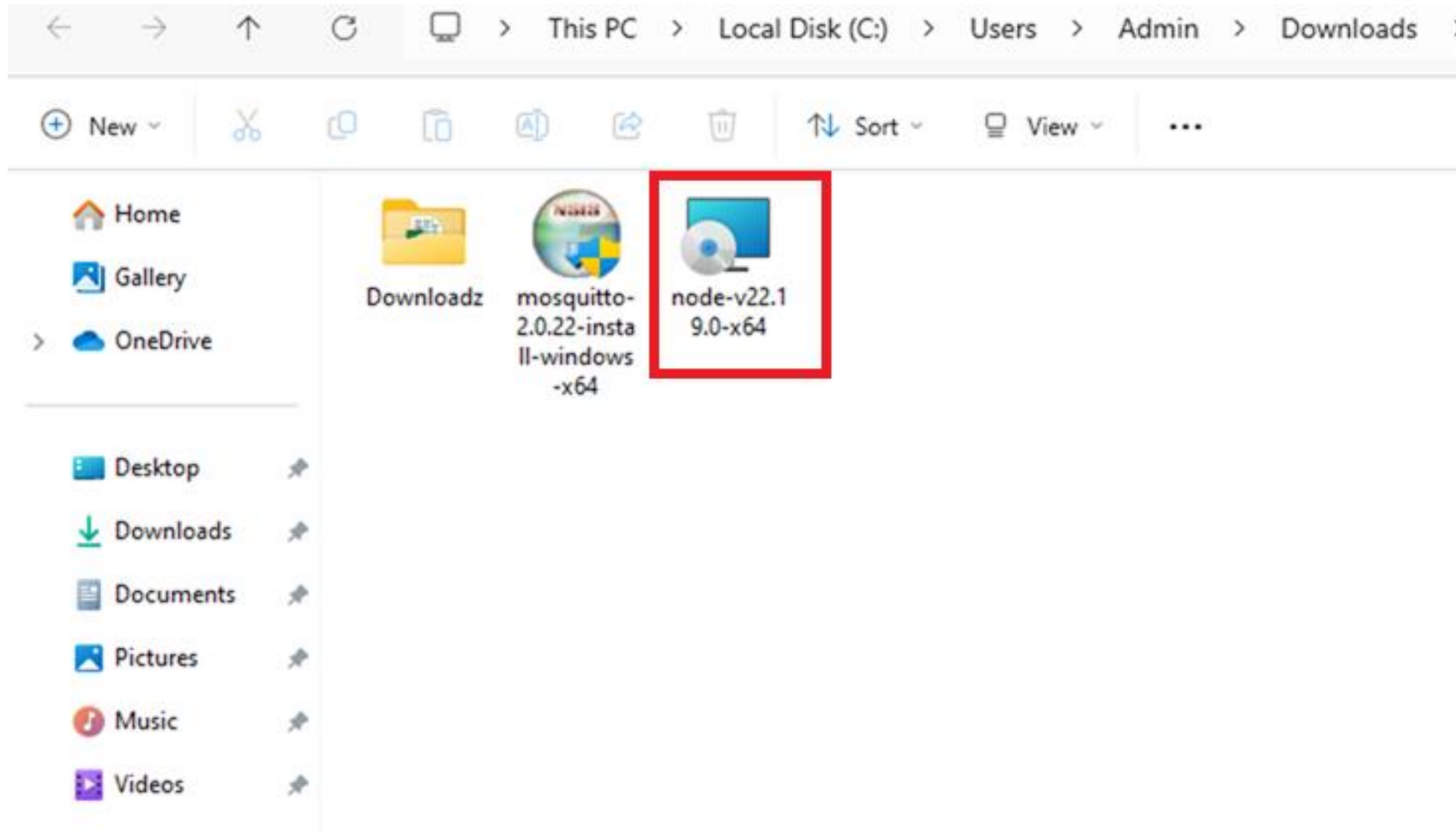
Standalone Binary (.zip)

Run the Installer

6. Click the **Install** button. You may see a User Account Control (UAC) prompt; click **Yes**.
7. Wait for the installation to complete. Click **Finish**.

Run the Installer

- Locate the downloaded file (e.g., node-v18.17.1-x64.msi) and double-click to run it.



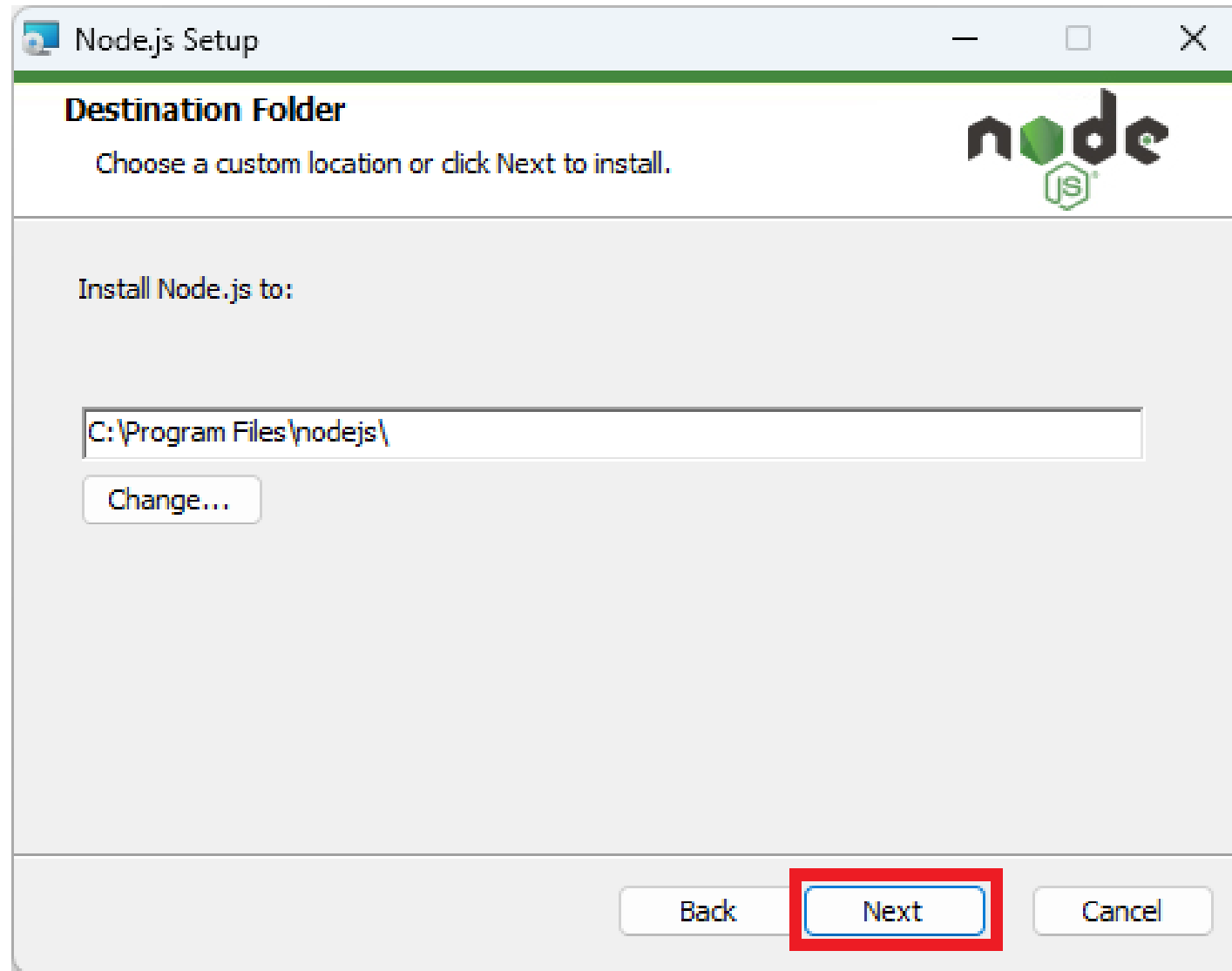
- The Node.js Setup Wizard will open. Click **Next**.



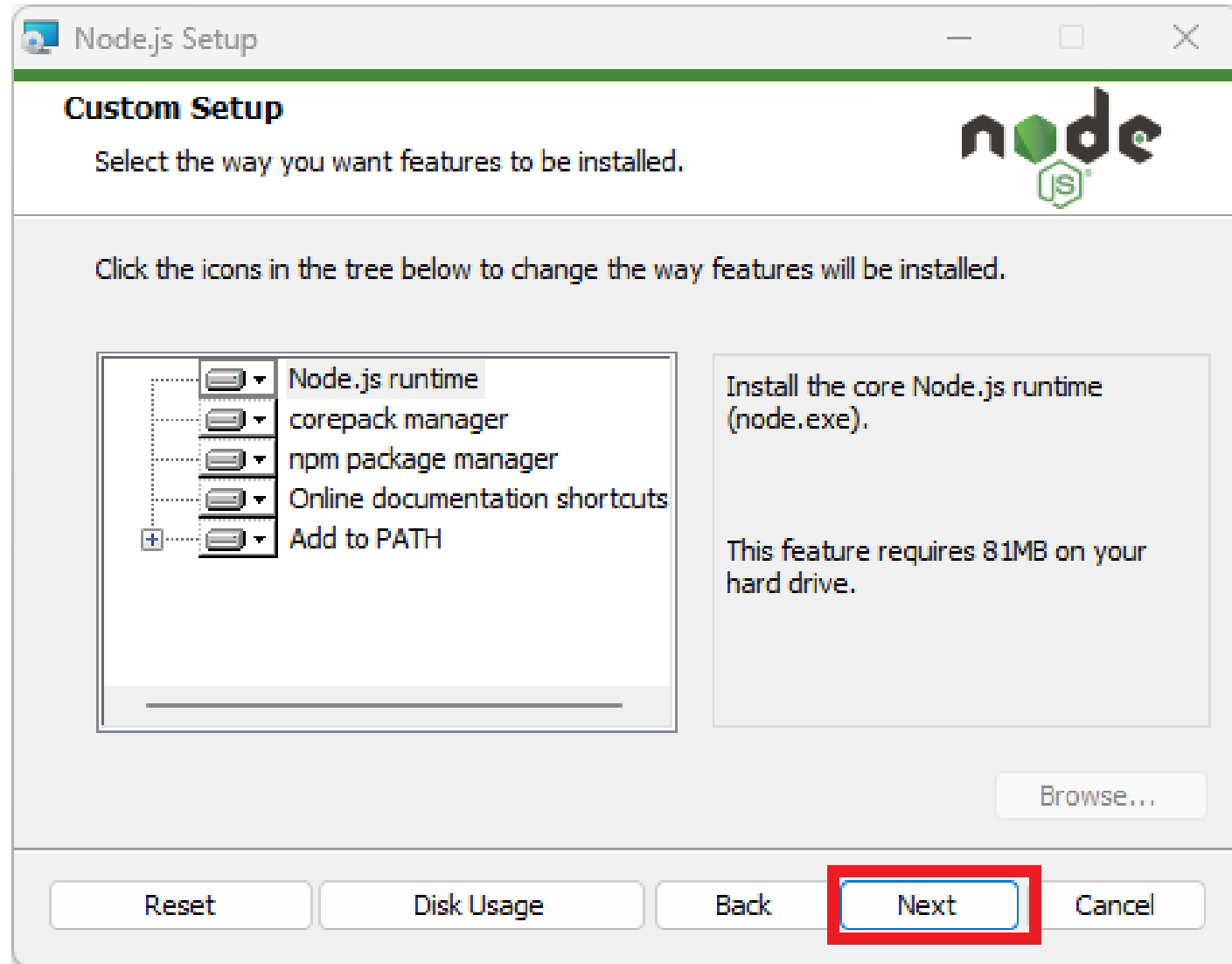
- **Accept the license agreement and click Next.**



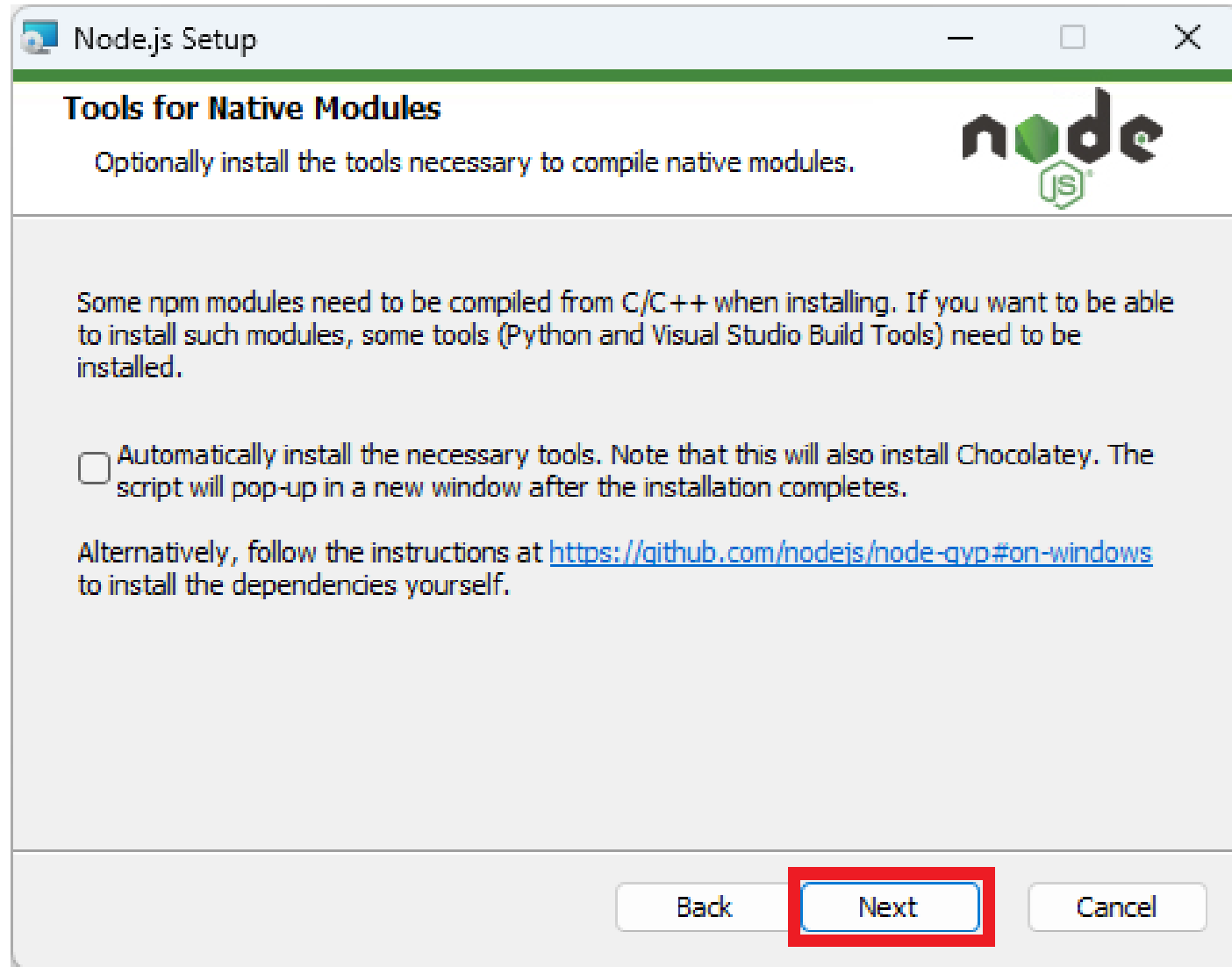
- **Choose the Installation Folder:** The default location is fine for most users (C:\Program Files\nodejs\). Click **Next**.



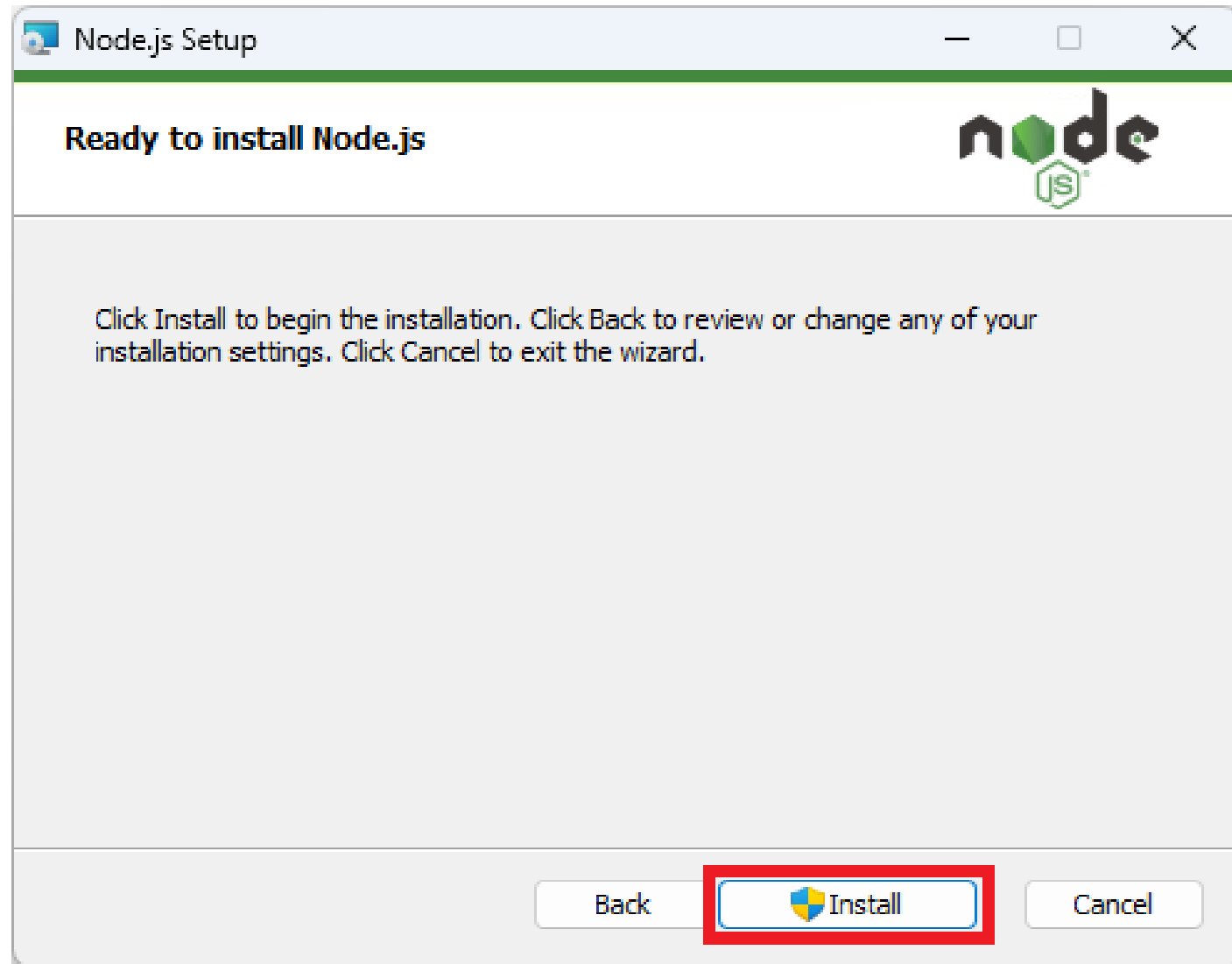
- **Customize the Setup:** The default settings on this page are perfect. It will:
 - Install Node.js core.
 - Install **npm** (the Node Package Manager, which is essential).
 - Automatically add Node.js and npm to your system **PATH**.
 - Download necessary tools (this option is fine to skip). Click **Next**.



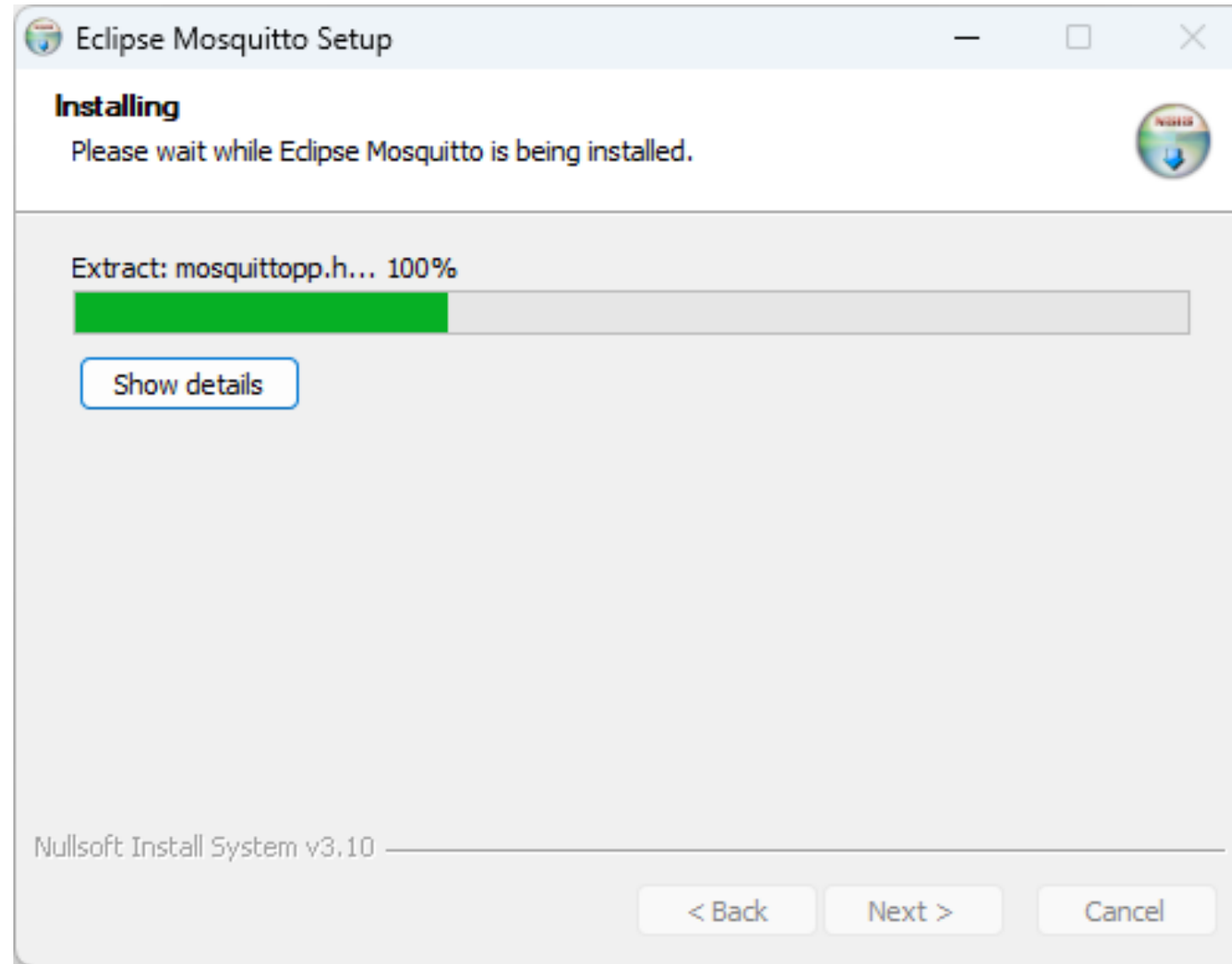
Select next. Don't click on the checkbox



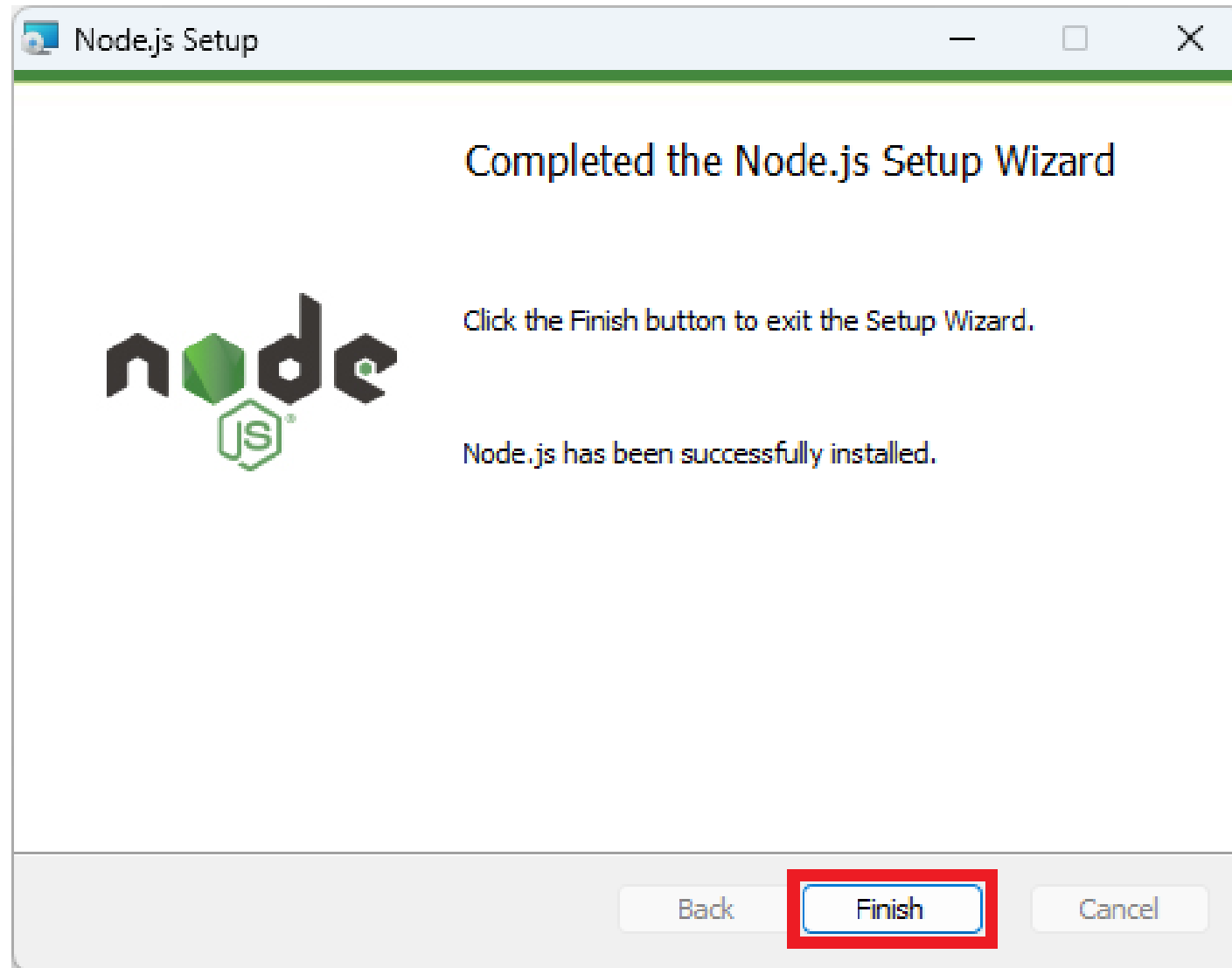
- Click the **Install** button. You may see a User Account Control (UAC) prompt; click **Yes**.



The installation is in progress.



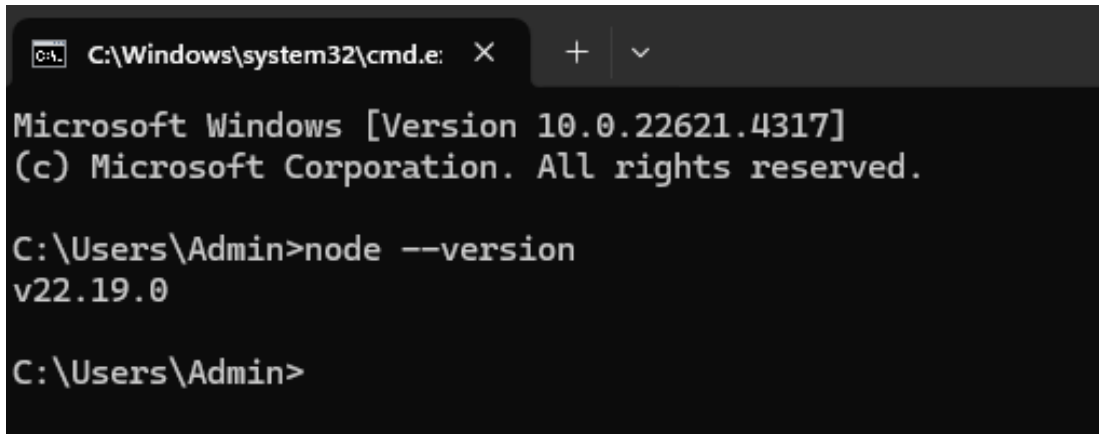
Wait for the installation to complete. Click **Finish**.



Verify the Installation

- Open the **Command Prompt**:
 - Press Win + R, type cmd, and press Enter.
 - Or, type "cmd" in the Windows Search bar and click on the Command Prompt app.
- In the Command Prompt window, type the following command and press Enter:

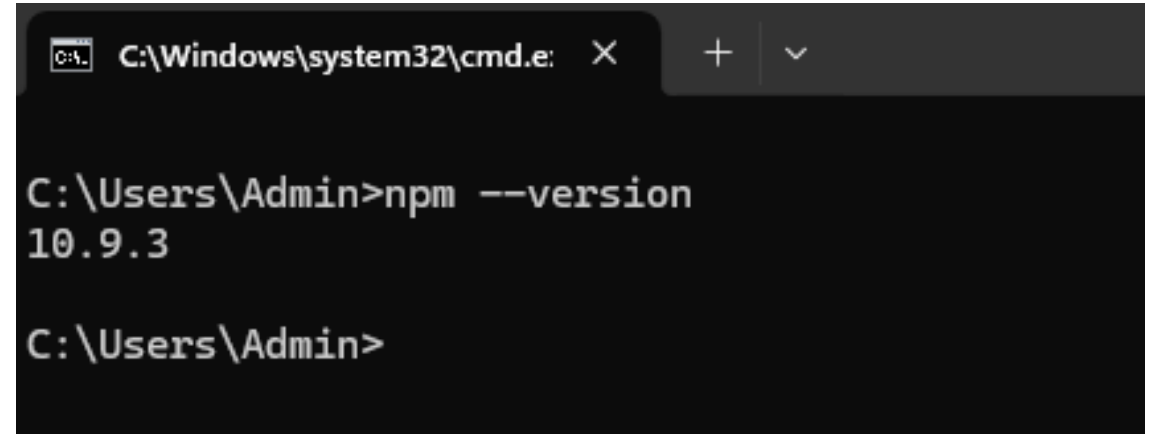
node --version
npm --version



```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>node --version
v22.19.0

C:\Users\Admin>
```



```
C:\Windows\system32\cmd.e: X + v
C:\Users\Admin>npm --version
10.9.3

C:\Users\Admin>
```


❑ Node-Red Installation

- Now, it is required to install Node-RED for Windows. Please go to the following site:
<https://nodered.org/docs/getting-started/windows>

■ Install Node-RED Globally

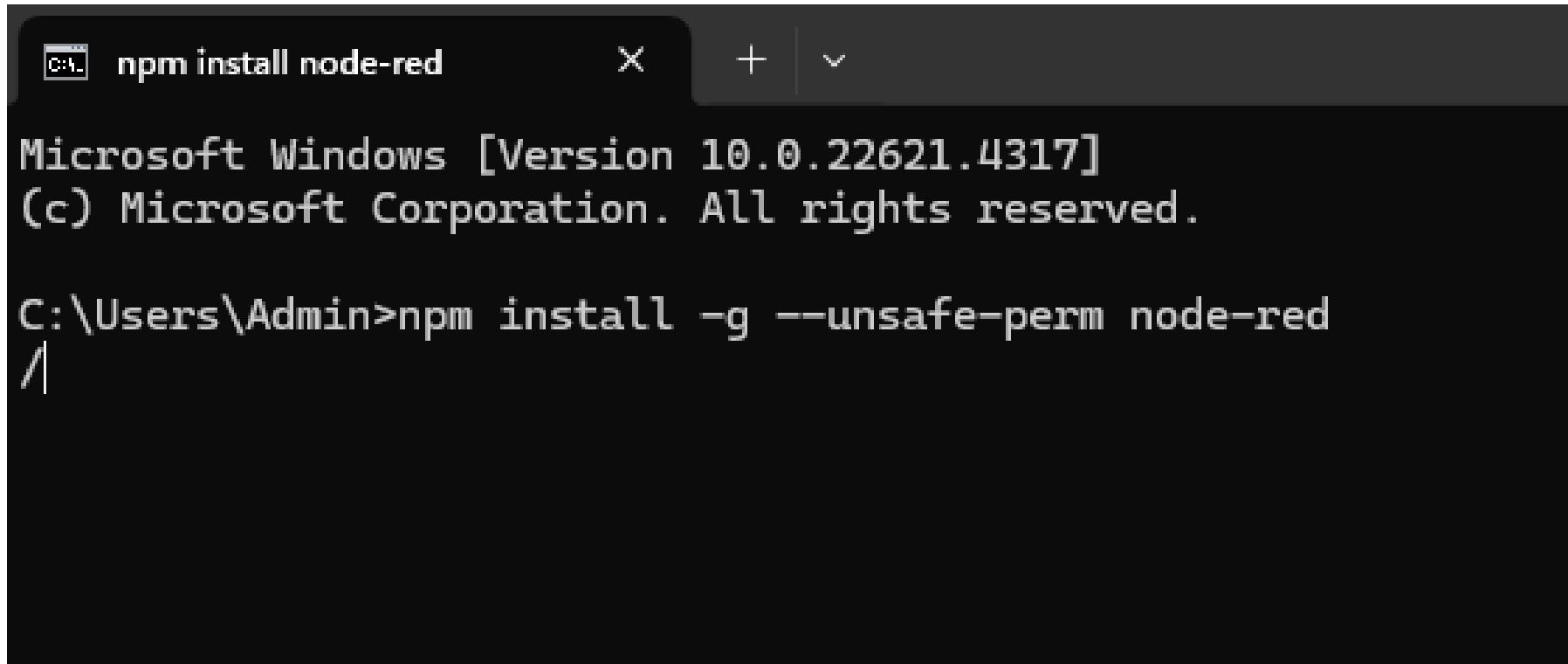
In the same Command Prompt window, type the following command and press Enter:

npm install -g --unsafe-perm node-red

- npm is the Node Package Manager, which you just installed.
- install tells npm to install a package.
- -g means "global", making the node-red command available from any folder on your system.
- --unsafe-perm is a flag often needed on Windows to avoid permission issues during installation.

- **Wait for the installation to complete.** npm will download and install Node-RED and all its dependencies.
- This may take a few minutes. You will see a lot of text scrolling by.

Installation is in progress.



```
npm install node-red
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>npm install -g --unsafe-perm node-red
/
```

```
C:\Windows\system32\cmd.e: X + v

Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>npm install -g --unsafe-perm node-red

added 276 packages in 2m

52 packages are looking for funding
  run `npm fund` for details
npm notice
npm notice New major version of npm available! 10.9.3 -> 11.6.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.0
npm notice To update run: npm install -g npm@11.6.0
npm notice

C:\Users\Admin>
```

- Once the installation is complete, close the Command Prompt.
- Then, open a new Command Prompt, type node-red, and press Enter.
- You will see a welcome message displayed

```

node-red
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>node-red
5 Sep 22:31:14 - [info]

Welcome to Node-RED
=====

5 Sep 22:31:14 - [info] Node-RED version: v4.1.0
5 Sep 22:31:14 - [info] Node.js version: v22.19.0
5 Sep 22:31:14 - [info] Windows_NT 10.0.22621 x64 LE
5 Sep 22:31:14 - [info] Loading palette nodes
5 Sep 22:31:15 - [info] Settings file : C:\Users\Admin\.node-red\settings.js
5 Sep 22:31:15 - [info] Context store : 'default' [module=memory]
5 Sep 22:31:15 - [info] User directory : \Users\Admin\.node-red
5 Sep 22:31:15 - [warn] Projects disabled : editorTheme.projects.enabled=false
5 Sep 22:31:15 - [info] Flows file : \Users\Admin\.node-red\flows.json
5 Sep 22:31:15 - [info] Creating new flow file
5 Sep 22:31:15 - [warn]
5 Sep 22:31:15 - [warn] Encrypted credentials not found
5 Sep 22:31:15 - [info] Server now running at http://127.0.0.1:1880/
5 Sep 22:31:15 - [info] Starting flows
5 Sep 22:31:15 - [info] Started flows

```

- <http://127.0.0.1:1880/>
- You will see a web page address displayed—copy it and paste it into your default browser.
- The Node-RED dashboard will appear, and it is now ready to use.

127.0.0.1:1880/#flow/074fe6f04d6335ba

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Node-RED

Deploy

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filter nodes

common

inject

debug

complete

catch

status

link in

link call

link out

comment

function

function

switch

change

range

template

delay

trigger

Flow 1

+

⌵

info

📄

📁

⚙️

⌵

Search flows

Flows

Flow 1

Subflows

Global Configuration Nodes

Flow 1

🔗

🔍

Flow

"074fe6f04d6335ba"

🔄

✕

ctrl-space

will toggle the view of this sidebar

Tektork Private Limited, Coimbatore

Add Dashboard and MQTT Nodes:

- To create the user interface, you'll need the Node-RED Dashboard.
- In the Node-RED editor, click the **hamburger menu (≡)** in the top right, select **Manage palette**, go to the **Install** tab, and search for **node-red-dashboard** and **node-red-contrib-mqtt-broker**. Install both.

Create the Flow:

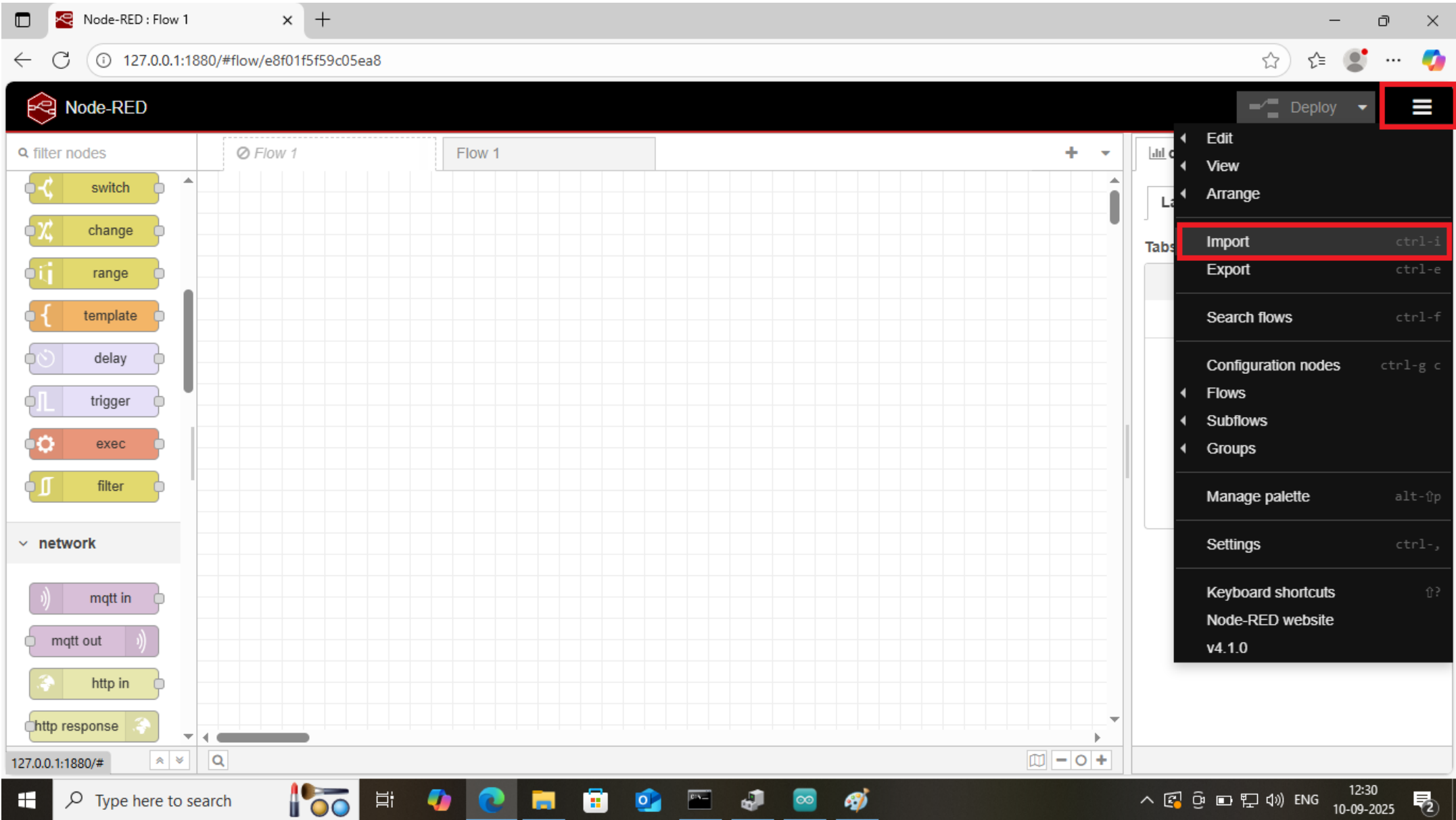
- Drag an **mqtt out** node from the palette onto the canvas. This node will publish messages to the MQTT broker.
- Double-click the **mqtt out** node. For the "Broker," click the edit pencil and configure it. Set the server to your computer's IP address (the same one you put in the ESP32 code) and the port to **1883**. The "Topic" should be esp32/led, matching what the ESP32 is subscribed to.
- Drag a **button** node from the dashboard section onto the canvas. This will be your "ON" switch. Double-click it and set the "Label" to "ON" and the "Payload" to **"ON"**.
- Drag a second **button** node. Configure it as the "OFF" switch with the "Label" as "OFF" and the "Payload" as **"OFF"**.
- Connect the output of both button nodes to the input of the **mqtt out** node.

Execution

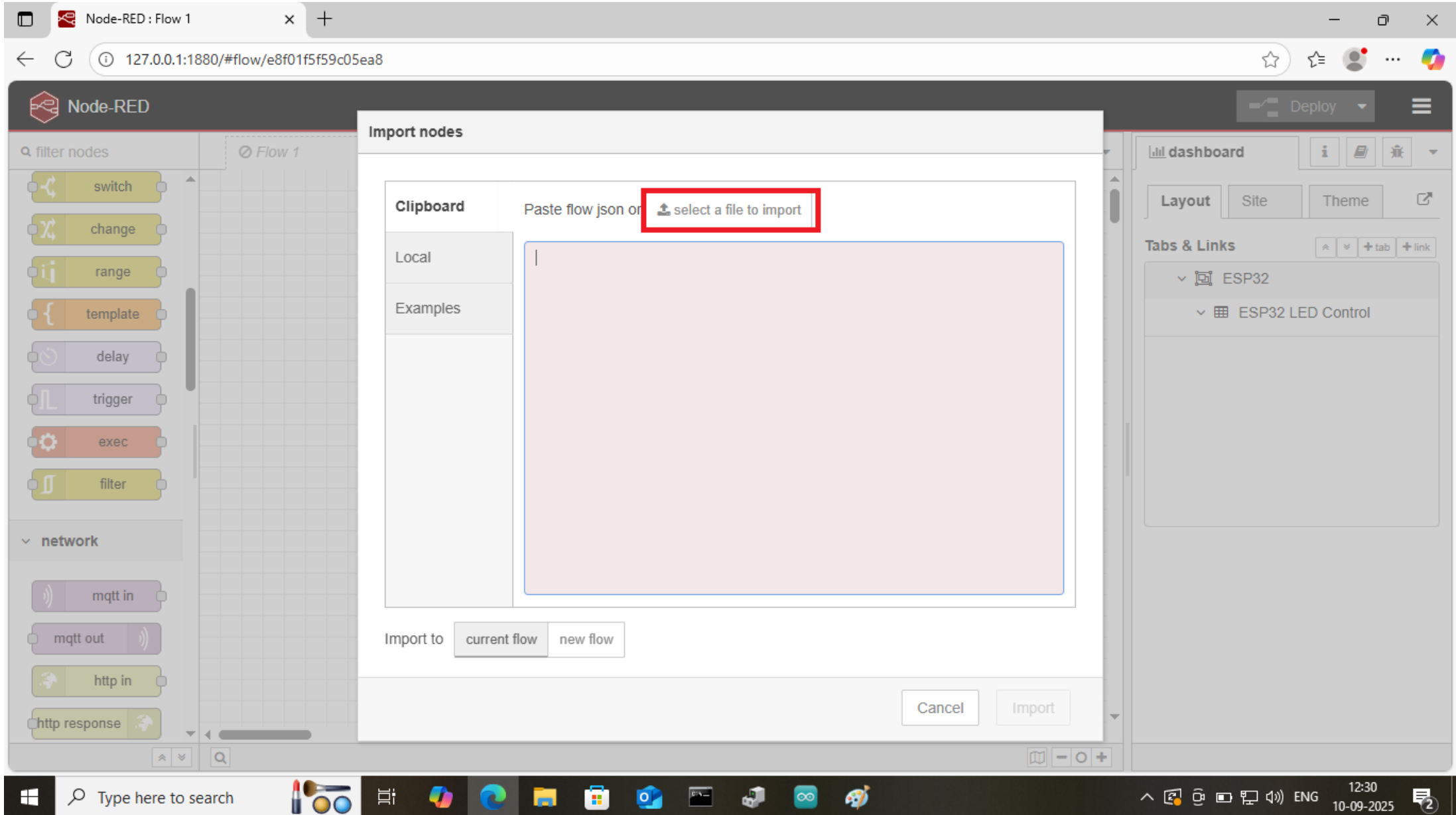
Now, you can test the setup.

- Ensure the ESP32 is powered on and connected to your Wi-Fi network.
- Start Node-RED from your terminal with the command `node-red`.
- Open the Node-RED dashboard by navigating to `http://YOUR_COMPUTER_IP:1880/ui` in your web browser. You will see the two buttons.
- Click the **ON** button, and the LED on your ESP32 should turn on. Click the **OFF** button, and it should turn off. This demonstrates the end-to-end communication from your web browser through Node-RED and the MQTT broker to your ESP32.

In the Node-RED editor, click the hamburger menu (≡) in the top-right corner and select 'Import'

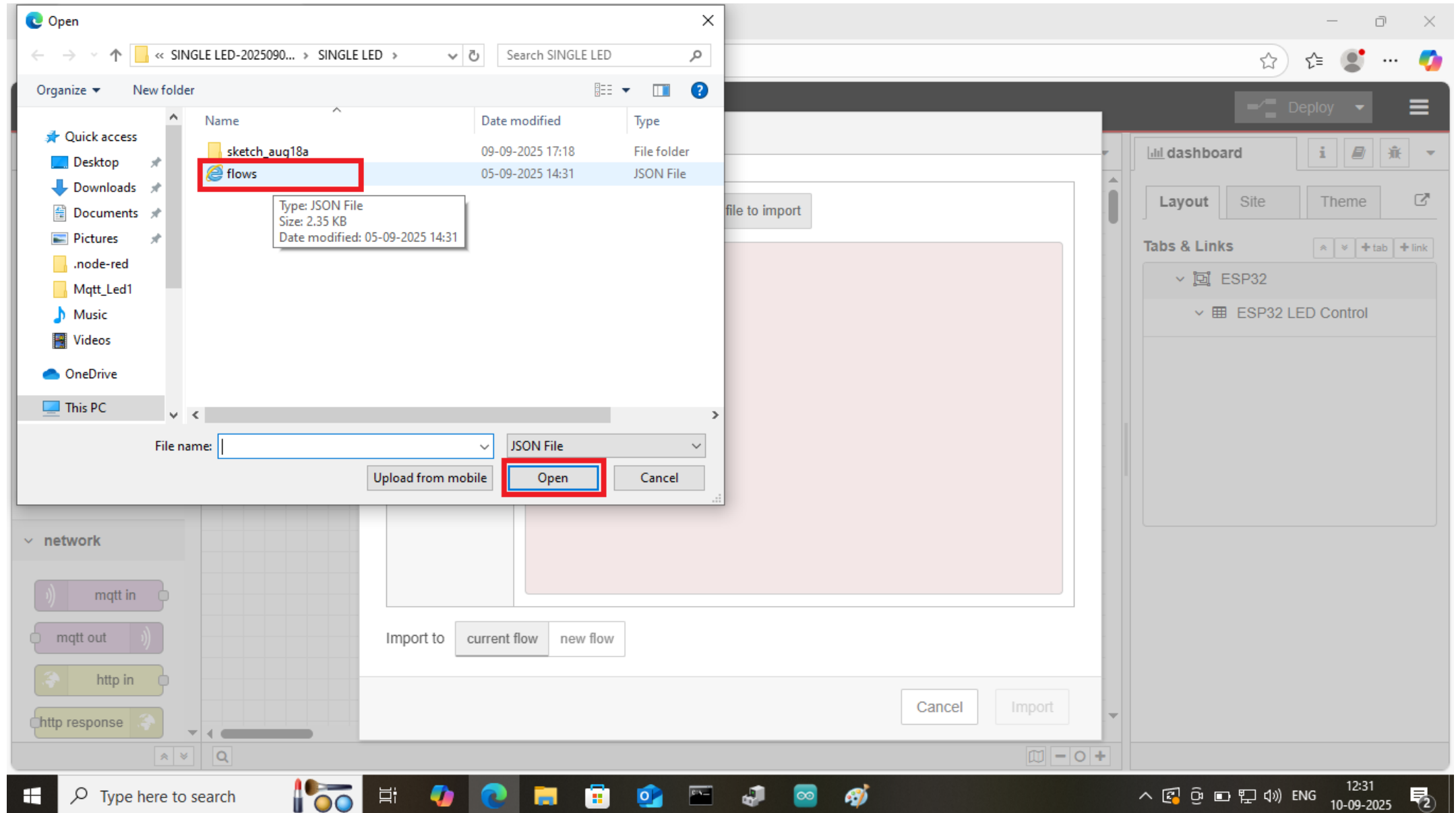


After selecting the 'Import' option, select the file you wish to import.



The screenshot shows the Node-RED web interface in a browser window. The address bar displays the URL `127.0.0.1:1880/#flow/e8f01f5f59c05ea8`. The main workspace is titled 'Node-RED' and shows a 'Flow 1' canvas. On the left, a sidebar lists various nodes under categories like 'filter nodes' and 'network'. An 'Import nodes' dialog box is open in the center, featuring a 'Clipboard' tab. This tab contains a text area for pasting flow JSON and a button labeled 'select a file to import', which is highlighted with a red rectangle. Below the text area, there are 'Import to' options: 'current flow' and 'new flow'. At the bottom of the dialog, there are 'Cancel' and 'Import' buttons. The right sidebar shows a 'dashboard' section with 'Layout', 'Site', and 'Theme' tabs, and a 'Tabs & Links' section listing 'ESP32' and 'ESP32 LED Control'. The Windows taskbar at the bottom includes a search bar and several application icons. The system clock in the bottom right corner shows the time as 12:30 on 10-09-2025.

After choosing the 'Import' option, browse to the flow file and select 'Open' to proceed.



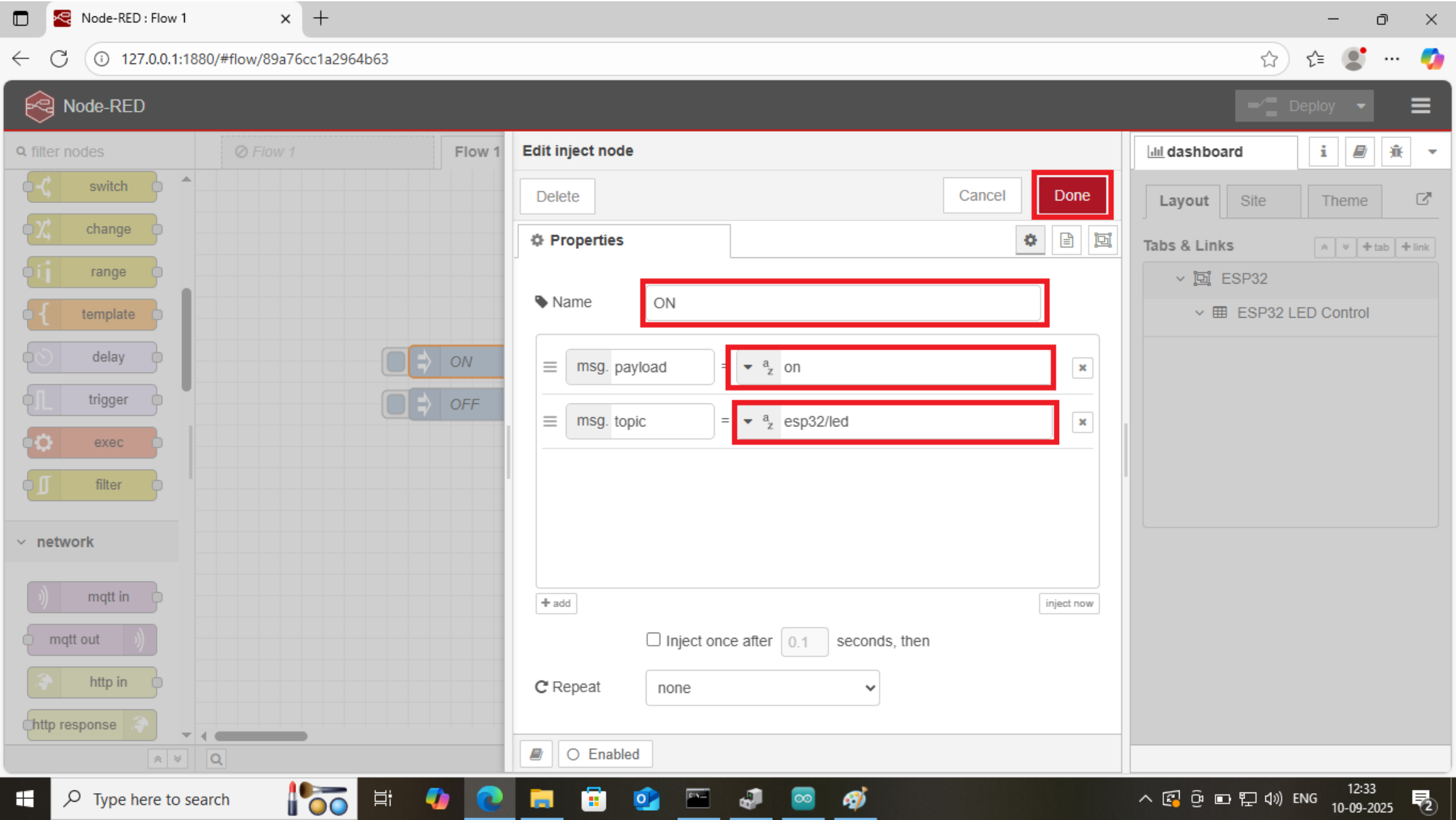
Once the file is open, click on the 'Import' option.

The screenshot shows the Node-RED web interface in a browser window. The address bar displays the URL `127.0.0.1:1880/#flow/e8f01f5f59c05ea8`. The main workspace is titled 'Node-RED' and shows a 'Flow 1' canvas. On the left, there is a 'filter nodes' sidebar with various nodes like 'switch', 'change', 'range', 'template', 'delay', 'trigger', 'exec', 'filter', and a 'network' section with 'mqtt in', 'mqtt out', 'http in', and 'http response'. A modal dialog titled 'Import nodes' is open in the center. It has a 'Clipboard' tab selected, showing a text area with the following JSON:


```
[
  {
    "id": "c4c67e4086e8d854",
    "type": "tab",
    "label": "Flow 1",
    "disabled": false,
    "info": "",
    "env": []
  },
  {
    "id": "inject_on",
    "type": "inject",
    "z": "c4c67e4086e8d854",
    "name": "ON",
    "props": [
      {
        "p": "payload"
      }
    ]
  }
]
```

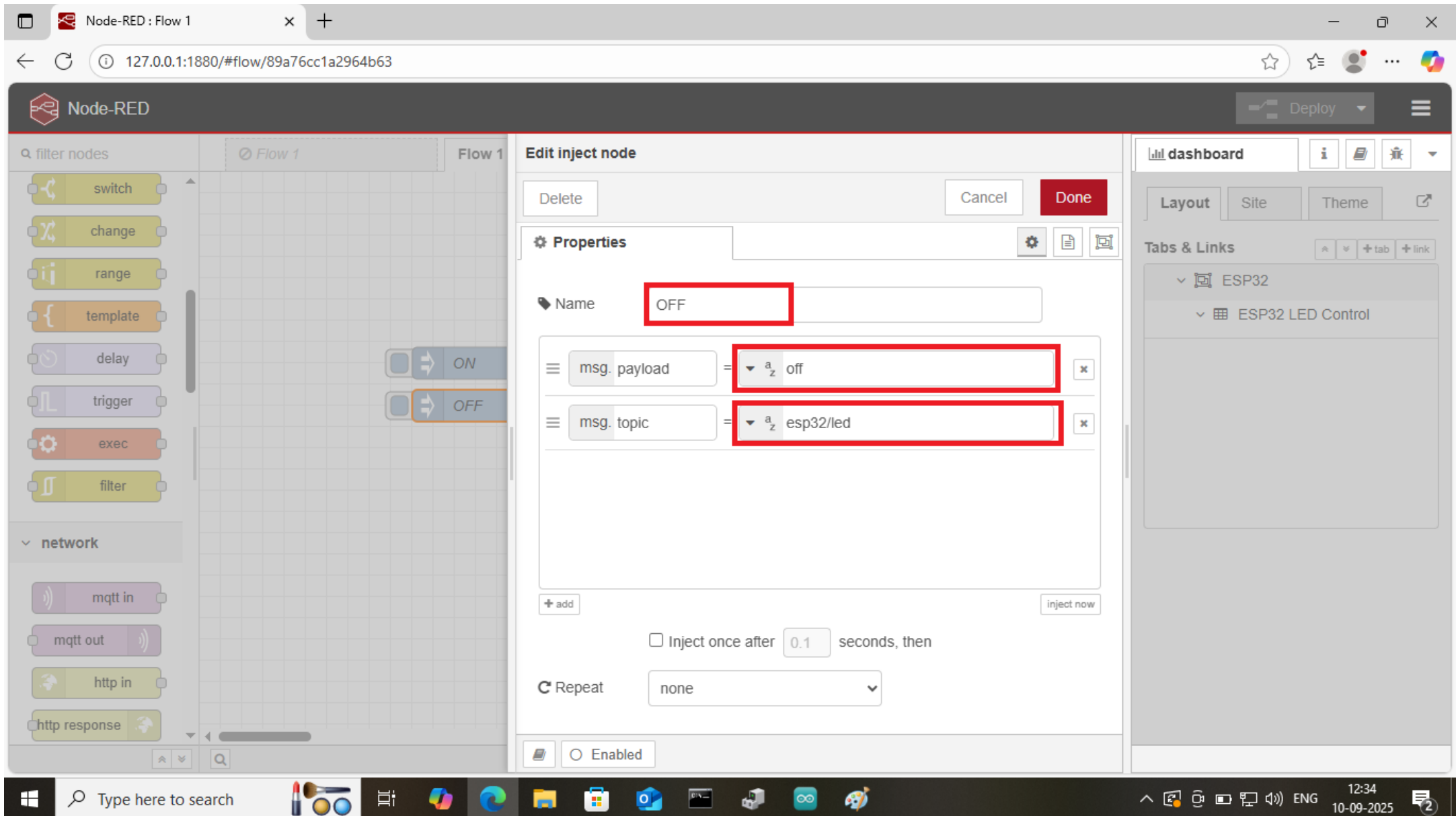
 A red box highlights the 'select a file to import' button above the text area. At the bottom of the dialog, there are 'Import to' buttons for 'current flow' and 'new flow', and a red box highlights the 'Import' button. The right sidebar shows a 'dashboard' section with 'Layout', 'Site', and 'Theme' tabs, and a 'Tabs & Links' section with 'ESP32' and 'ESP32 LED Control' tabs. The Windows taskbar is visible at the bottom with the search bar and various application icons. The system clock shows 12:32 on 10-09-2025.

Next, double-click the first node to edit it, input the necessary settings, and select 'Done'.



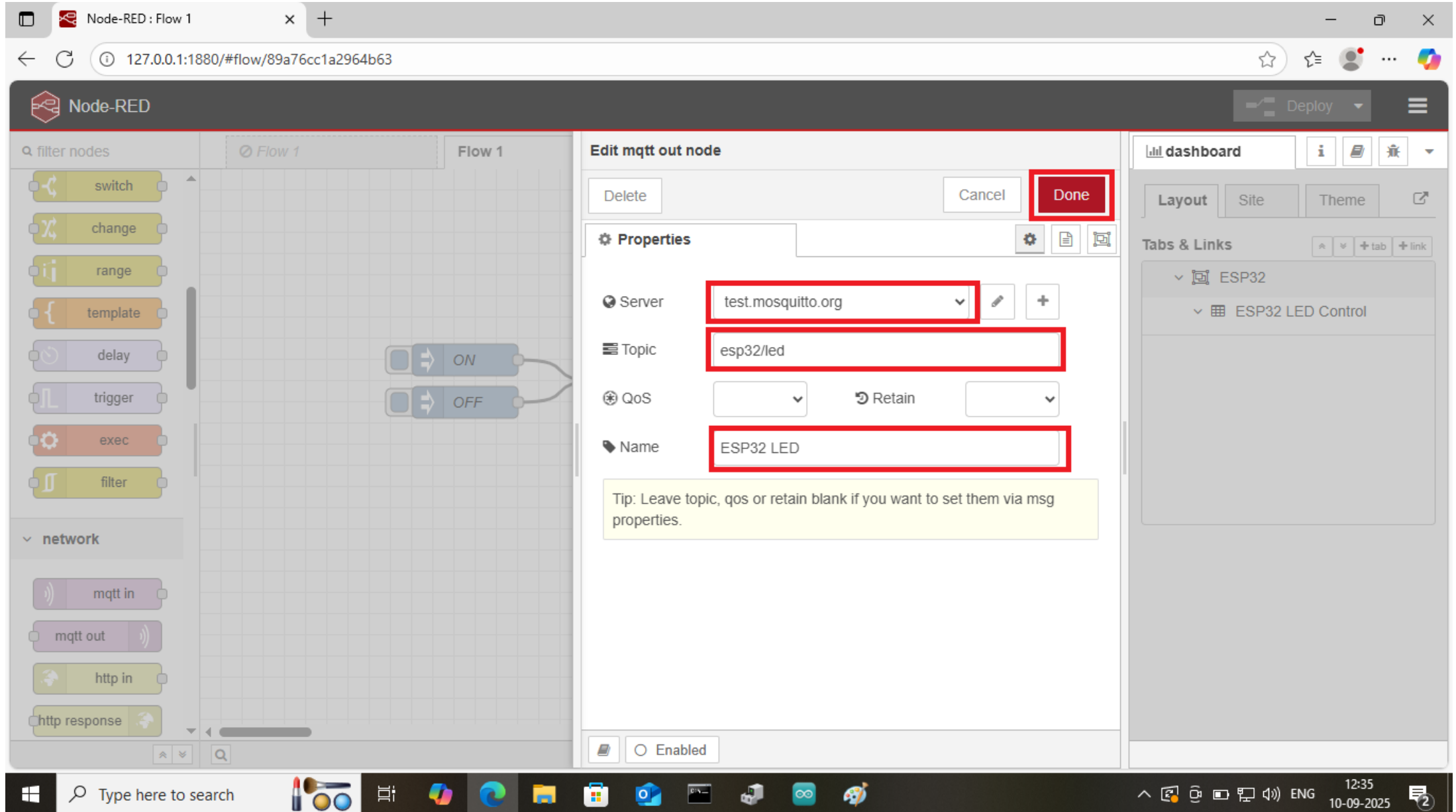
The screenshot shows the Node-RED web interface in a browser. The main workspace displays a flow with two inject nodes labeled 'ON' and 'OFF'. The 'ON' node is selected, and its configuration panel is open on the right. The configuration panel has a 'Done' button highlighted with a red box. Below the 'Properties' section, the 'Name' field is set to 'ON' and is also highlighted with a red box. The 'msg. payload' field is set to 'on' and the 'msg. topic' field is set to 'esp32/led', both highlighted with red boxes. The 'Inject once after' field is set to '0.1' seconds, and the 'Repeat' dropdown is set to 'none'. The 'Enabled' checkbox is checked. The left sidebar shows various nodes, and the right sidebar shows the dashboard with tabs for 'ESP32' and 'ESP32 LED Control'. The Windows taskbar at the bottom shows the time as 12:33 on 10-09-2025.

Next, double-click the second node to edit it, input the necessary configuration, and select 'Done' to save the changes.



The screenshot displays the Node-RED web interface in a browser window. The main workspace shows a flow with two inject nodes. The second node is selected, and its configuration panel is open. In the 'Edit inject node' panel, the 'Name' field is set to 'OFF'. The 'msg. payload' field is set to 'off', and the 'msg. topic' field is set to 'esp32/led'. The 'Inject once after' field is set to '0.1' seconds, and the 'Repeat' dropdown is set to 'none'. The 'Enabled' checkbox is checked. The right sidebar shows a dashboard with a tab for 'ESP32' and a sub-tab for 'ESP32 LED Control'. The bottom of the screen shows the Windows taskbar with the search bar and various application icons.

Next, double-click the third node (mqtt) to edit it, provide the necessary configuration, and click 'Done' to save your changes

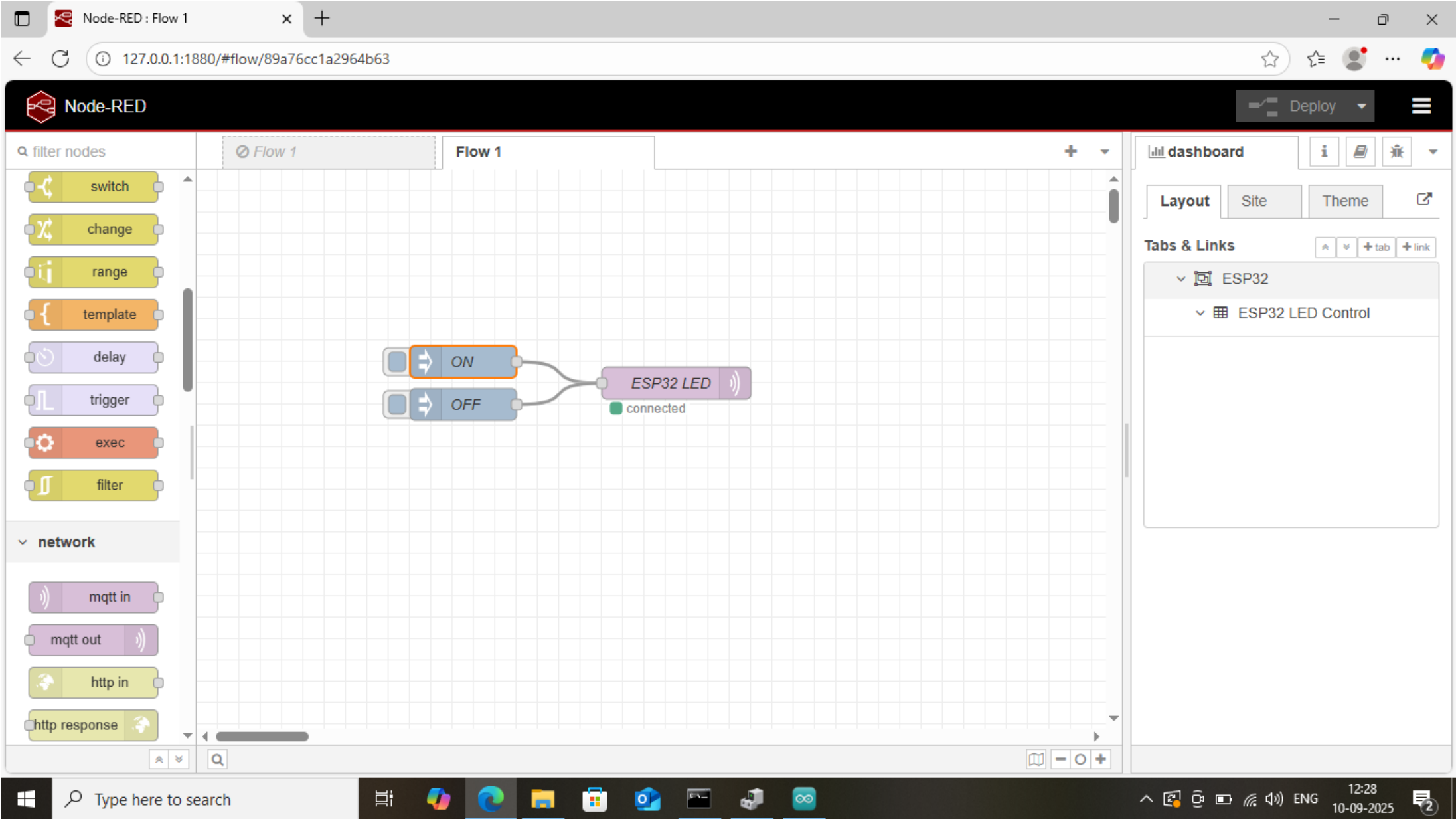


The screenshot displays the Node-RED web interface in a browser window. The main workspace shows a flow with two nodes: 'ON' and 'OFF', both connected to an 'mqtt out' node. The 'mqtt out' node is selected, and its configuration panel is open on the right. The configuration panel is titled 'Edit mqtt out node' and includes a 'Done' button highlighted with a red box. The 'Properties' section contains the following fields:

- Server:** test.mosquitto.org (highlighted with a red box)
- Topic:** esp32/led (highlighted with a red box)
- QoS:** 0 (dropdown menu)
- Retain:** false (checkbox)
- Name:** ESP32 LED (highlighted with a red box)

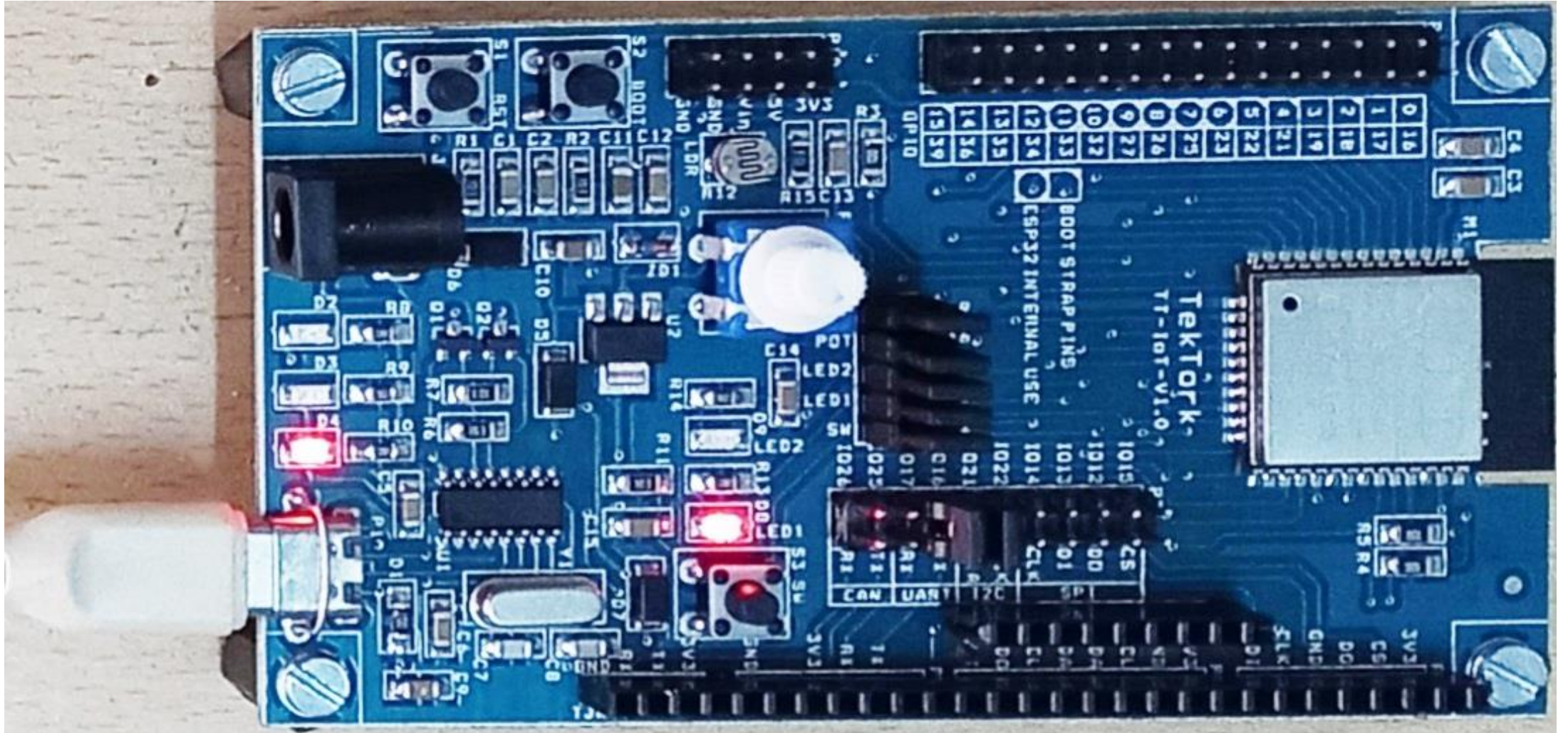
A tip message is displayed below the fields: "Tip: Leave topic, qos or retain blank if you want to set them via msg properties." The 'Enabled' checkbox is checked. The left sidebar shows the 'network' category with nodes like 'mqtt in', 'mqtt out', 'http in', and 'http response'. The right sidebar shows the 'dashboard' section with a 'Layout' tab and a 'Site' tab. The bottom of the screen shows the Windows taskbar with the search bar and various application icons.

Finally, click the 'Deploy' button to connect the nodes to the MQTT server. You can now control the LED on the ESP32 using the Node-RED flow



The screenshot displays the Node-RED web interface in a browser window. The address bar shows the URL `127.0.0.1:1880/#flow/89a76cc1a2964b63`. The interface includes a left sidebar with a 'filter nodes' search bar and a list of nodes categorized into 'network' and other types. The 'network' category is expanded, showing 'mqtt in', 'mqtt out', 'http in', and 'http response' nodes. The main workspace, titled 'Flow 1', contains a flow with two 'mqtt in' nodes labeled 'ON' and 'OFF' connected to an 'ESP32 LED' node. The 'ESP32 LED' node has a green indicator and the text 'connected'. The right sidebar shows a 'dashboard' section with 'Layout', 'Site', and 'Theme' tabs, and a 'Tabs & Links' section with a tree view showing 'ESP32' and 'ESP32 LED Control'. A 'Deploy' button is visible in the top right corner of the interface. The Windows taskbar at the bottom shows the time as 12:28 on 10-09-2025.

Output



Thank You